



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

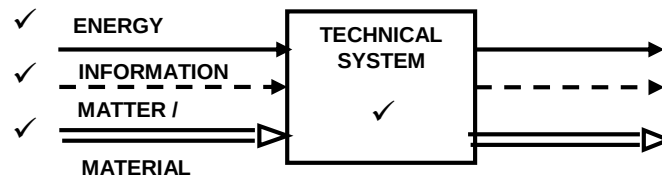
**MECHATRONICS SYSTEMS
NQF LEVEL 2**

XX February 2021

This marking guideline consists of 6 pages.

QUESTION 1

1.1



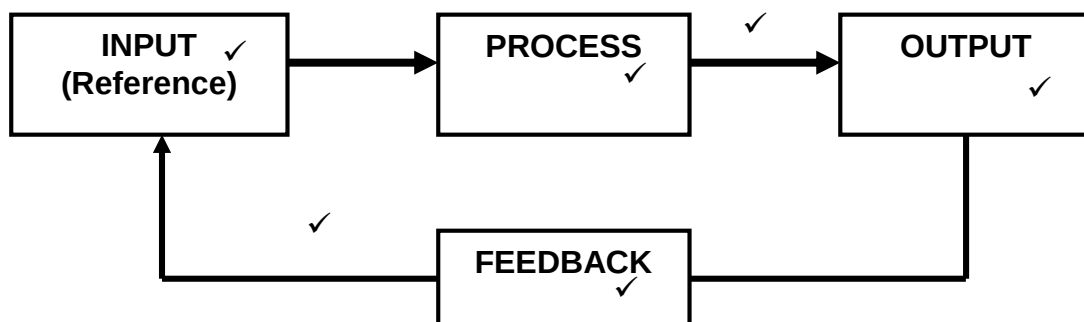
(4 × 1) (4)

1.2

- The energy flow and related conversion.
- The information flow.
- The material flow.
- The operator panel and related components.
- The machine/application and related components.
- The control panel and related components
- Feedback status sensors indicated.
- Open or closed loop control can be determined.

(Any other relevant information 8 × 1) (8)

1.3



(6 × 1) (6)

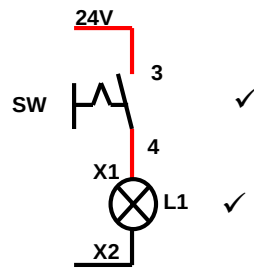
1.4

The process of changing one form of energy into another, such as nuclear energy into heat or solar energy into electrical energy.

(2)
[20]

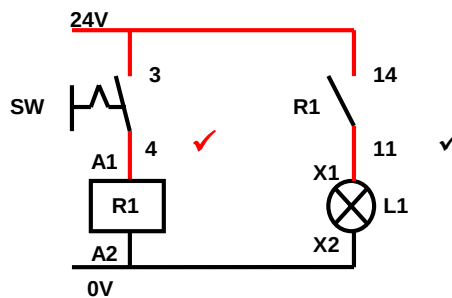
QUESTION 2

2.1



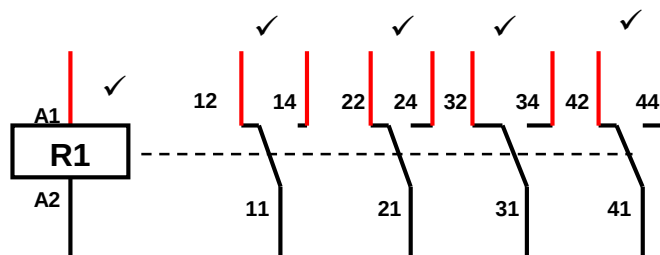
(2)

2.2



(2)

2.3



ONE mark, if all terminals are correctly labelled✓

(6)

2.4 Remote control is when the operator sits and monitors and manages the processes ✓ a distance away from the actual work area. ✓

(2)

2.5 Safety voltage refers to a voltage that is considered to be a low voltage (1–50V) ✓ that will not shock a person under normal conditions is used in control circuits. ✓

(2)

2.6 Intrinsic safety is the region where a flash over arc will not cause a fire or explosion ✓ in an area where chemicals are used. ✓

(2)

- 2.7
- Indication lamp
 - Horn
 - Siren
 - Buzzer
 - Dial
 - Gauge

(Any 3 × 1)

(3)

2.8 Indication lamp is commonly used.

(1)

[20]

QUESTION 3: Air generation

- Air distribution
- Air application

(3)

3.2 PVC tubing

(1)

3.3 Pneumatic valve

(1)

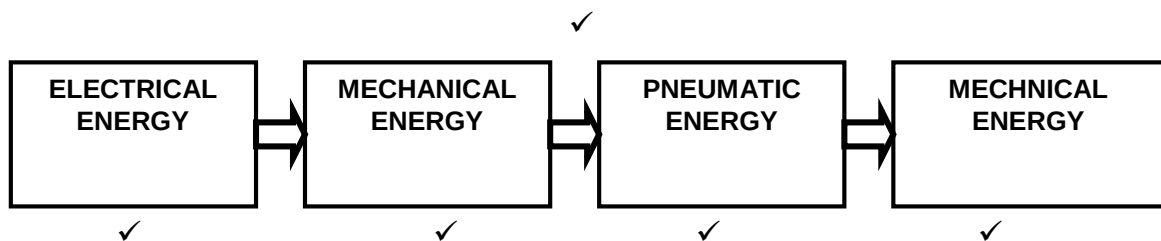
3.4 The spool by virtue of its position will control the flow of air in the application.

(2)

- 3.5
- Put the system into zero pressure.
 - Reset all cylinders into the initial positions.
 - Stop all cylinders in their current position.
 - Combine two or more abovementioned options.

(4)

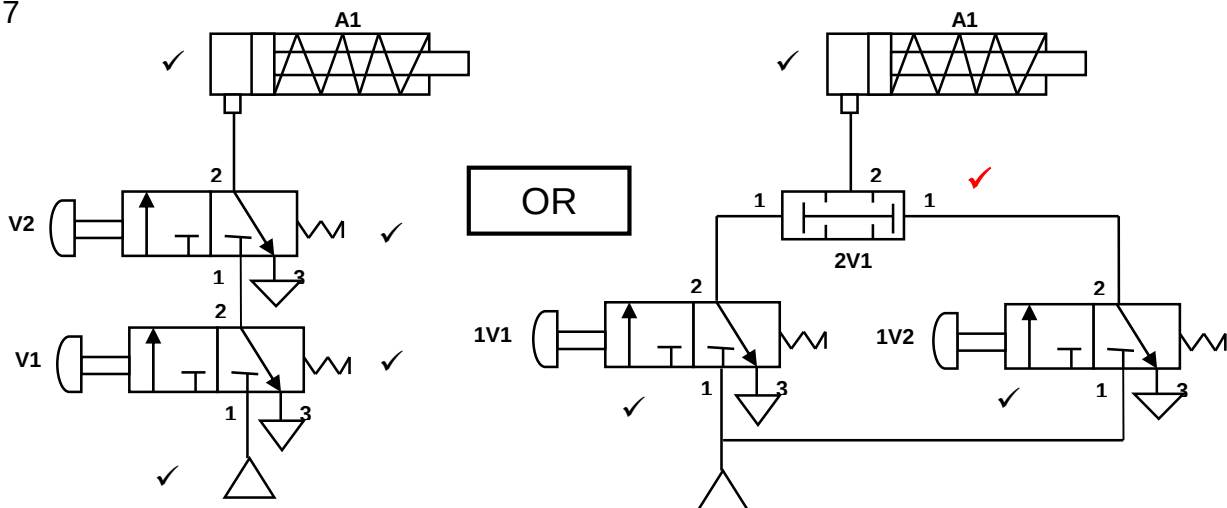
3.6



(5 × 1)

(5)

3.7



(Direct control or indirect control is acceptable 4 × 1)

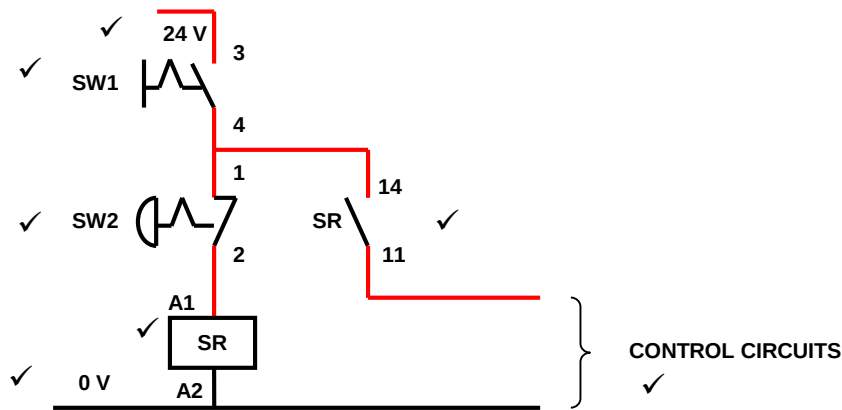
(4)

[20]

QUESTION 4

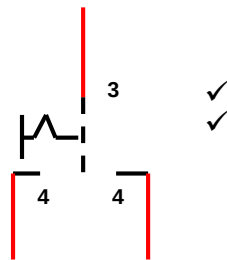
- 4.1 Emergency stops are used to prevent injury or death to workers or damage to machinery when a fault condition or hazardous condition exists✓ to ensure that the mechatronic system must be reset after the emergency stop has been activated✓ as required by legislation.✓ (3)

4.2



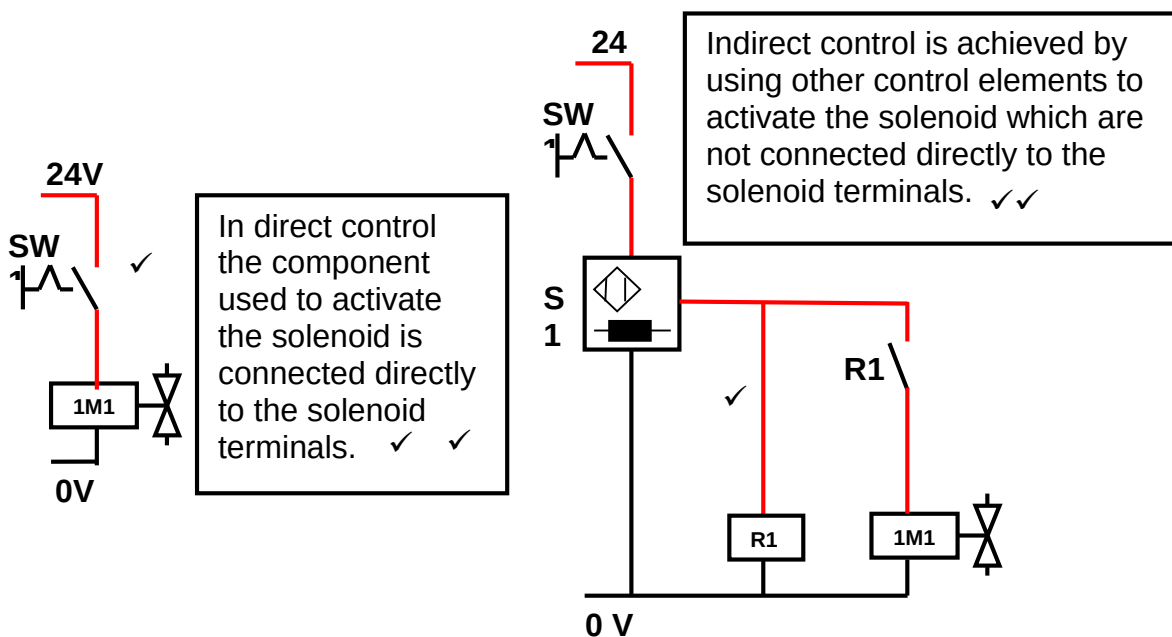
(7)

4.3



(2)

4.4



(6)

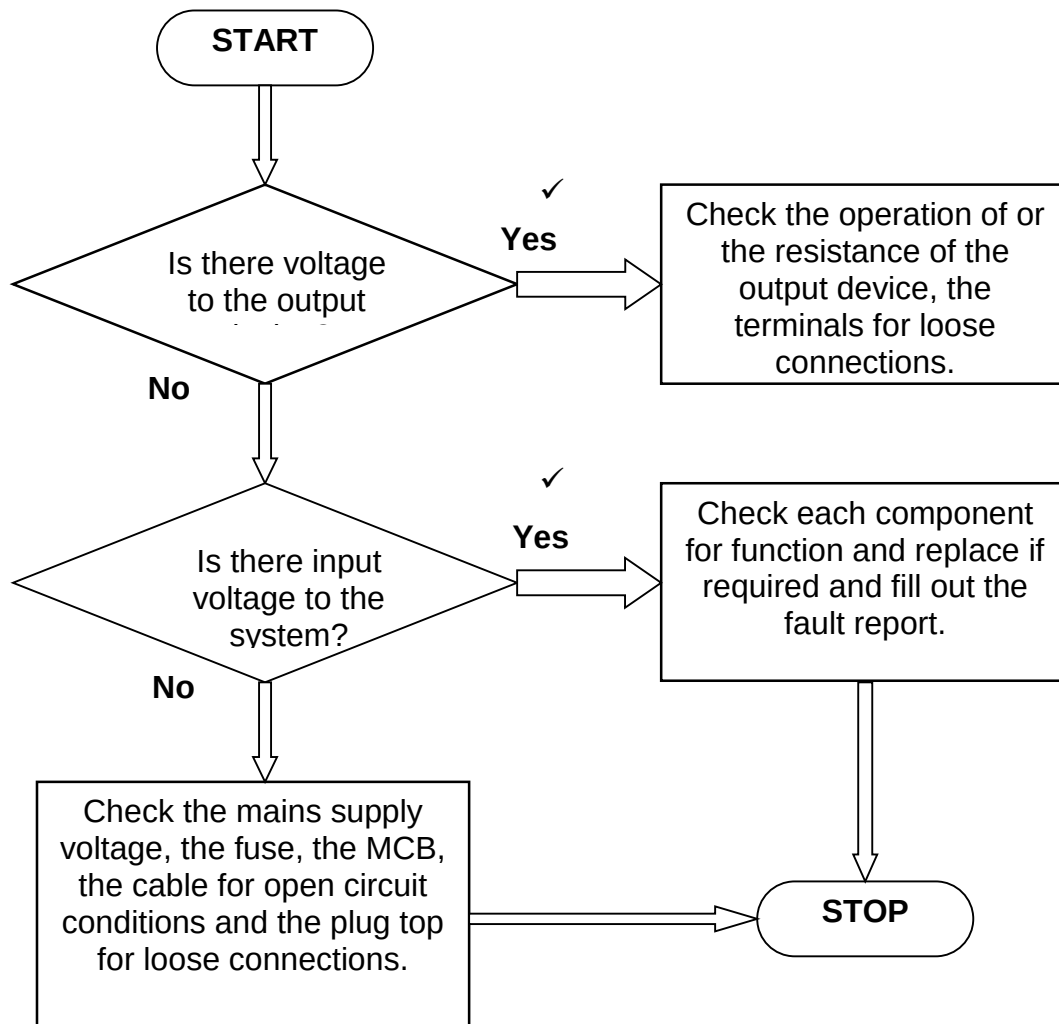
- 4.5 $I = V/R$ ✓
 $= 24/48$
 $= 0,5 \text{ A}$ ✓ (unit must be shown to get the mark)

(2)
[20]

QUESTION 5

- 5.1
- Check if water is coming out of the nozzle (output).
 - Check if water is coming out the tap (input).
 - Check along the hose pipe for cracks, breaks, kinks, blockages, etc. (through the circuit).
- (3)

5.2



(9)

- 5.3 The sole purpose of the mechatronic person is to ensure that fault elimination is carried out✓ with the minimum production down time✓ to maintain production.✓
- (3)

- 5.4 Ensure that the supply is switched off✓ and locked out.✓
- (2)

- 5.5
- Blocked lines
 - Seizure of units
 - Breakages
 - Leakages
 - Dropping of air pressure
 - Incorrect switching
- (Any relevant faults 3 × 1)
- (3)

[20]**TOTAL: 100**